

2-Mark Questions & Answers — MPCA Unit 3: The Intel Microprocessor 8086 Architecture

Q1. List the functional units of the 8086 microprocessor.

The 8086 CPU is divided into two independent functional units:

1. **Bus Interface Unit (BIU):** Manages instruction prefetching, memory bus communication, segment registers, and physical address generation.
2. **Execution Unit (EU):** Decodes micro-instructions, executes arithmetic/logical operations in ALU, and manages general-purpose registers and flags.

Q2. What is memory segmentation in 8086?

Memory Segmentation is a technique that divides the 1MB physical memory space into 64KB logical blocks called segments (Code Segment, Data Segment, Stack Segment, Extra Segment). It allows a 16-bit processor to address 20-bit physical addresses using segment base registers and offset registers.

Q3. Define minimum mode and maximum mode of 8086.

- **Minimum Mode (MN/MX' = 1):** Single-processor system mode where 8086 generates all bus control signals (**RD** , **WR** , **ALE** , **M/IO**) directly.
- **Maximum Mode (MN/MX' = 0):** Multi-processor system mode where bus control signals are generated by an external Bus Controller chip (Intel 8288).

Q4. What is the purpose of demultiplexing the address/data bus?

The 8086 uses multiplexed pins **AD0 - AD15** to save physical pin count. **Demultiplexing** separates the 20-bit memory address from 16-bit data using octal latches (74LS373) driven by the Address Latch Enable (**ALE**) signal.

Q5. State the size of the 8086 instruction queue.

The 8086 Bus Interface Unit (BIU) contains a **6-byte Instruction Queue** that prefetches instruction bytes sequentially from memory when the EU is busy, enabling instruction pipelined prefetching.